Your grade: 92%

Your latest: **92%** • Your highest: **92%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. In which scenarios could reinforcement learning-based adversarial attacks prove to be effective? (Select all that Apply.) **0.8 / 1 point**

- [x] Causing image classification models to misclassify objects through pixel alterations

  This should not be selected

  Incorrect. Pixel alteration attacks on image classification models typically involve adversarial examples generated from neural networks, not reinforcement learning.

- [ ] Generating incoherent text from natural language processing models
- [x] Directing robotic control systems to execute unsafe maneuvers

  Nice work

  Correct! Adversarial attacks using reinforcement learning can cause robotic systems to take unsafe actions, bypassing standard safety protocols.

- [x] Influencing game-playing AI to lose deliberately

  Nice work

  Correct! Reinforcement learning-based adversarial attacks can manipulate game-playing AI, guiding them to make suboptimal moves or deliberately lose.

- [x] Steering autonomous driving systems towards incorrect actions

  Nice work

  Correct! Reinforcement learning-based adversarial attacks can influence autonomous driving systems by manipulating their decision-making processes, causing them to take dangerous or incorrect actions.

2. How do reinforcement learning-based adversarial attacks generally deceive the learning agent? **1 / 1 point**

- ( ) By adjusting the loss function to distort the training process
- (•) By modifying the reward structure to misdirect the agent's policy choices
- ( ) By introducing noise into the training data to disrupt the learning process
- ( ) By altering the activation functions within the neural network

Nice work

Correct! These attacks often alter the reward outcomes to mislead the agent's learning and decision-making processes.

3. Which statement most accurately characterizes reinforcement learning-based adversarial attacks? **1 / 1 point**

- ( ) They involve adding noise to the input data to deceive the neural network.
- ( ) They involve modifying the network architecture to integrate adversarial layers.
- (•) They involve manipulating the reward mechanism to mislead the learning agent.
- ( ) They involve using backpropagation gradients to inject adversarial influences.

Nice work

Correct! These attacks focus on altering the reward system to misguide the learning agent's behavior.

4. Which tactics are employed in reinforcement learning-based adversarial attacks? (Select all that Apply.) **0.8 / 1 point**

- [ ] Exploiting gradients during the backpropagation process
- [ ] Altering the network's architecture
- [x] Modifying the state transition dynamics

  Nice work

  Correct! Adjusting how states transition can be a strategy used in reinforcement learning-based adversarial attacks.

- [x] Manipulating the reward mechanism

  Nice work

  Correct! One common tactic in these attacks is to alter the reward system to affect the agent's decisions.

- [x] Introducing adversarial perturbations into the input data

  This should not be selected

  Incorrect. This tactic is associated with neural network-based attacks, not reinforcement learning-based ones.

5. What is a principal distinction between reinforcement learning-based attacks and neural network-based attacks? **1 / 1 point**

- ( ) Reinforcement learning-based attacks do not involve a learning phase, unlike neural network-based attacks.
- ( ) Reinforcement learning-based attacks are geared towards supervised learning, whereas neural network-based attacks are not.
- ( ) Reinforcement learning-based attacks are invariably more intricate than neural network-based attacks.
- (•) Reinforcement learning-based attacks focus on manipulating the reward system, while neural network-based attacks target the input data.

Nice work

Correct! The principal distinction is that reinforcement learning-based attacks manipulate the reward structure to drive decision-making in adversarial scenarios, while neural network-based attacks generally target the input data to deceive the model.